PAPER_668: UQFF Primordial Black Hole Stability Analysis

Subtitle: Mass classification of PBHs under UQFF: stable_DM / marginal / evaporating. UQFF minimum DM-viable mass derived. **Module:** UQFFStabilityPrimordialBH
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Abstract

We systematically classify primordial black holes under UQFF by their evaporation prospects: stable dark matter, marginal, or evaporating. The minimum DM-viable mass shifts significantly compared to standard Hawking theory.

1. Standard Evaporation Timescale

$$\tau_{std} = \frac{5120\pi G^2 M^3}{\hbar c^4}$$

Standard boundary: $\tau\_std = t\_H$ (Hubble time) → $M\_boundary \approx 5\times10^{11}$ kg.

2. UQFF Timescale

$$\tau_{UQFF} = \frac{\tau_{std}}{1 - f_{TRZ}} \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot \exp\left(\frac{U_m}{k_B T_H}\right)$$

Factor ≈ 30×.

3. Mass Classifications

Category	Condition	Mass range (standard)	Mass range (UQFF)
stable_DM	$\tau\_UQFF \geq t\_H$	$M \geq 5\times10^{11}$ kg	$M \geq 1.6\times10^{11}$ kg
marginal	$0.1\ t\_H \leq \tau < t\_H$	narrow window	wider window
evaporating	$\tau < 0.1\ t\_H$	$M < 4\times10^{11}$ kg	$M < 1.3\times10^{11}$ kg

4. Numerical ($M = 10^{12}$ kg)

- $\tau\_std \approx 10^{10}$ yr
- $\tau\_UQFF \approx 3\times10^{11}$ yr **> Hubble time** → viable DM

5. C++ Module

`UQFFStabilityPrimordialBH.h / .cpp` — Session 172 CP4 #252 — `UQFFStabilityPrimordialBHCalculator`

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